The Longevity–Labor Connection:

Do High-Life-Expectancy Counties Keep Older Adults in the Workforce?

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Abstract

This paper plans to examine whether county-level life expectancy predicts the labor force participation rate and the employment rate among older adults in the United States. The data comes from IPUMS USA and the Global Health Data Exchange from 2009–2019. I estimate four separate fixed-effects regression models for individuals aged 55–65 and 66–74, while controlling for various demographic and economic characteristics. Ultimately, the results find that life expectancy is negatively associated with both employment rate and labor force participation among individuals aged 66-74. However, there was no statistically significant relationship for those aged 55–65, suggesting that health plays a limited role in determining labor supply before traditional retirement ages. These findings indicate that greater longevity does not necessarily correspond to extended labor force participation or employment. Instead, higher life expectancy appears to be associated with longer periods spent in retirement, with potential implications for the fiscal sustainability of Social Security and related public programs.

1 Introduction

Life expectancy has been on an exponential rise since the 1970s, increasing from just below 71 to nearly 79 prior to COVID-19. With this exponential increase, more people are able to work at older ages. Healthcare plays a particularly unique role when it comes to the field of economics. As the overall spending on health can typically show how well we as a society can perform economically. One big economic factor related to healthcare is the process of retirement. Retirement is specifically important to understand because if retirement is not planned for correctly or done within a certain window, serious consequences can set in (homelessness, social security dependence, etc). Unfortunately, in today's landscape, young voters are ill-informed regarding the process and procedure of retirement and social security. This paper hopes to educate the overall public, but more specifically, younger voters, so that they can be well-informed and can support and understand the implications of policies and reforms that will be beneficial for them later in life.

This paper examines whether county-level life expectancy predicts labor force participation or employment rates among individuals aged 55 and older. I use a fixed effects OLS regression with five covariates to account for demographic and economic differences between counties. To provide a more nuanced analysis, I examine the age groups 55–65 and 66–74 separately to represent two distinct phases of later life. The results indicate that life expectancy predicts labor force participation and employment rates among adults ages 66–74, but not among those ages 55–65. More specifically, the findings suggest a negative marginal effect, implying that higher life expectancy is associated with lower labor force participation and employment among older adults. These results challenge the assumption that longer life necessarily translates into longer working lives. Instead, they suggest that increases in longevity may extend the period of retirement, raising important implications for the sustainability of programs such as Social Security as more individuals live longer while spending fewer years in the labor force.

2 Background

The question I am looking to answer has been studied in some respects but in different capacities and focusing on different aspects that can affect older workers. To get a better understanding of the previous

research that has already been conducted on the topic of older people in the work force. The first study I looked at was from Andrew Aitken. In his study, Aitken examines how changes in life expectancy have affected the workforce in collaboration with the Organization for Economic Co-operation and Development (OECD). Aitken analyzes trends within the labor force of OECD-affiliated countries within the age groups 16-29, 30-54, and 55-64. Aitken comments that the longer life expectancy of humans has led to increased participation of older workers in the labor force. Aitken views this as a huge benefit, as it has increased workers' longevity and therefore contributed to higher employment rates across OECD countries. However, the primary concern he addresses is that employment policies have not changed with the market to support longer working lives. Aitken finds through his research that many participants in the labor force retire before they are eligible for their pensions after age 60 which is due to several factors, with the biggest being the increase in the average age of labor-market exit. Aitken concludes that for the economy to adjust to this increase in the number of older workers, "it requires a comprehensive approach that reflects the profound demographic, economic, and technological transformations reshaping labor markets." (Aitken 2025). The main takeaway that we can take from this research is the fact that many workers still retire before reaching pension eligibility. While we can assume this is applicable to the United States, which my research focuses on, it is still important to note that this issue is apparent in other economies.

This second article by Kalwij and Vermeulen looks at a question that more closely relates to mine in comparison to the previous research. In this paper, the authors look at the labor force participation of older individuals in eleven different European countries, and the issue of potential endogenous self-reported health, as a decline in health is often cited as a key determinant of the decision to retire. The authors address this issue by comparing self-reported and objective health indicators across eleven different countries, using data from the Survey of Health, Ageing and Retirement in Europe (SHARE). The authors treat the potential false self-reported health as an omitted variable problem, and look at whether incorporating the objective indicators help explain labor participation models. They found that some countries do a good job of being accurate in their self-reporting, and therefore controlling for objective health indicators does not add a lot to the analysis. However, there are other countries where it is the opposite; objective health indicators have a significant effect on the analysis and that there is bias in self-reported health. Their results highlight both the multi-dimensional nature of health and the necessity of controlling for objective health indicators when analyzing health and labor force participation. From this paper it is clear health plays an important role in retirement decisions across European countries as well as the fact that self-reported health data can be potentially unreliable and misleading.

The final article that I examined was by Michael V. Leonesio, Benjamin Bridges, Robert Gesumaria, and Linda Del Bene. This paper showed how dramatically labor force participation among older Americans has changed in recent years. The authors explain that labor force participation among older workers was lower from 1900 to 1980 due to how easy it was to retire early (due to increased wealth, Social Security, etc.). However, during the mid-80s, these benefits started to be questioned as there seemed to be too many retirees and not enough workers in the labor force. After this period of panic, an increase in full-time employment not only raised the earnings for older workers but also eased the pressure on the Social Security system. The authors conclude by explaining that these occur because of the Age Discrimination in Employment Act, which protects workers and job applicants who are 40 years of age or older from age-based discrimination.

These claims also add to my argument, as this study shows that labor force participation among older Americans has risen since the 1980s. It also backs my findings that employment growth among older adults helps relieve pressure on Social Security.

3 Contribution

The prior literature that we have looked at examines the relationship between health and labor supply, primarily at the international and national levels. It has been concluded that there is a strong relationship between the two of them. However, there is little work that examines whether geographic variation in life expectancy (specifically within the United States) predicts labor force behavior at a local level, particularly across counties. This study contributes to the literature by combining county-level health measures with individual labor outcomes while also trying to confirm the claims made in previous literature. In addition, the use of fixed effects for the counties allows the isolation of individual county changes over time, helping to control for unobserved characteristics such as economic conditions, access to healthcare infrastructure, and local culture. This provides a more precise estimate of how improvements in local health relate to labor force participation as individuals approach and exceed traditional retirement ages.

4 Data and Empirical Design

For this project, I used four separate data sources to construct our final data set. I chose to select the sample over a 10-year period (2009-2019) so the sample would be large enough while also minimizing the effects of COVID-19 in our analysis. I also chose to look at data on the county level to get a more in-depth understanding of the labor force over different geographical and political areas. The largest part of the dataset we collected was via IPUMS USA. IPUMS is a database that provides integrated census and survey data from around the world to facilitate research. It offers large collections of harmonized microdata, making it easier to study social and economic trends across different countries and time periods. Using the IPUMS database, we were able to create three of our five control variables. The control variables from IPMUS are share_BA (percentage of a county with a BA degree or higher), share_female (proportion of women in the county) , and share_white (ratio of white people in the county). The second part of the dataset is from the Global Health Data Exchange (GHDx). We used their United States Healthy Life Expectancy data, which was collected via the Institute for Health Metrics and Evaluation at the University of Washington. The data set gives life expectancy by county, race, and ethnicity from 2009 to 2019. The variable for life expectancy in the dataset is represented as Avg_Life_Expec. It is important to note that the average life expectancy is measured by years left to live and not an actual age (If the person is 59 and expected to live to 80, his life expectancy would be 21). The third data source we used was from the Kaiser Family Foundation. KFF provided all of the state-level data for our Government Healthcare spending. This data measures the amount spent on healthcare per capita across all U.S. states. The fourth and final source used in creating the complete data set was from Yahoo Finance. They provide all of the data for the controlled variable State_Retirement_Age, which measures the average retirement age across all U.S states. These variables will help in our evaluation and analysis of the economic and labor trends within each county so we can compare them to one another. The figures below showcase some visual insights and trends of the relevant control

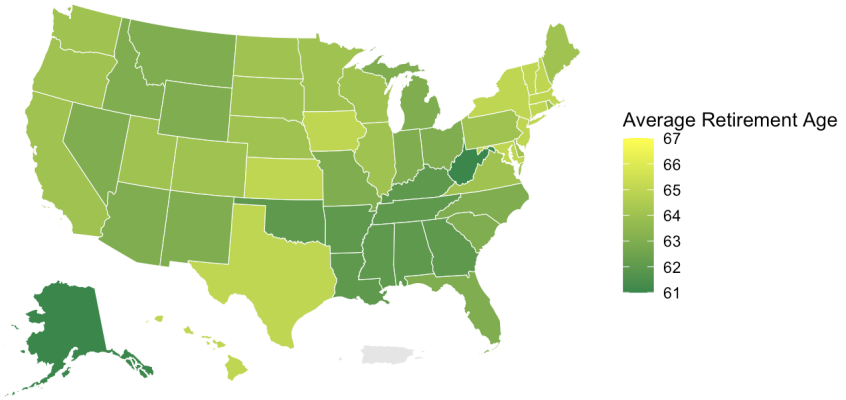

Figure 1: Retirement Age by State (2019)

variables.

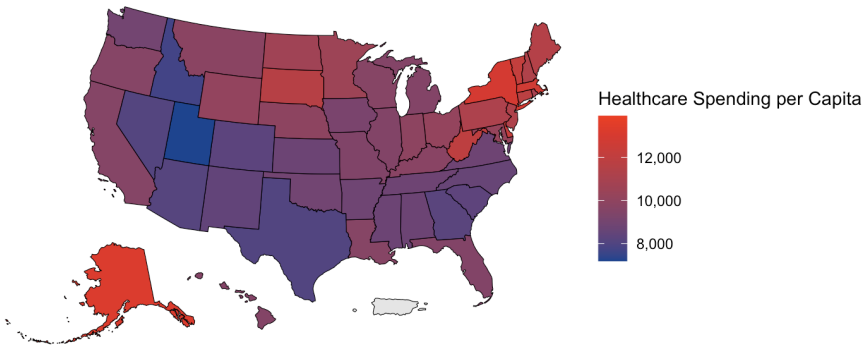

Figure 2: Government Healthcare Spending per Capita (2019)

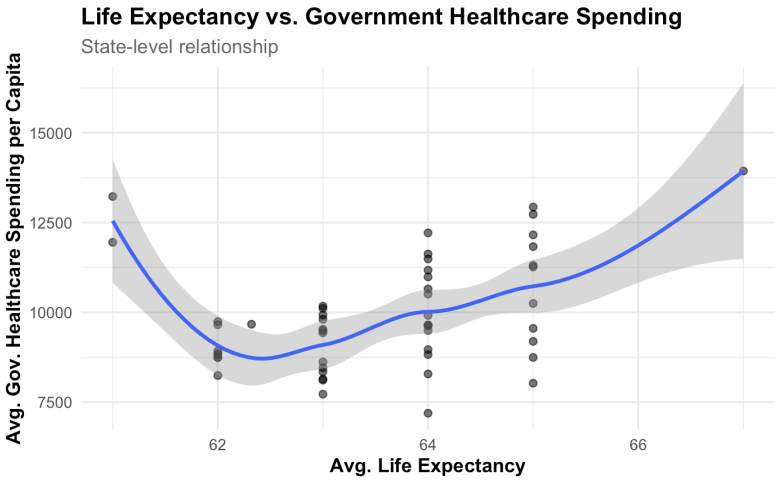

Figure 3: Scatterplot of Life Expectancy and Government Healthcare Spending

For the main econometric methodology, I ran 4 estimated fixed-effects ordinary least squares regression models (FEOLS) to control for unobserved county differences and to accommodate the large data set. These regressions were run on life expectancy and employment rate for our two age groups: 55-65 and 66-74. I used AvgLifeExpec, share_bachelors, share_female, and share_white, Gov_Health_Spending, and StateRetirementAge to predict labor force participation rate and employment rate for the different age groups. The models for 55-65 and 66-74 were evaluated separately to identify differences in labor market behavior around typical retirement ages. Life expectancy affects pre-retirement and post-retirement workers differently, and I wanted to see if increased life expectancy in higher life expectancy counties results in a higher labor force participation or employment rate.

Based on economic theory, I predict a strong relationship between health and labor supply over the life cycle, because individuals factor their health into their decisions of how long to work and what kind of work they do. There are many situations in which health is a major predictor of how long an individual stays in the workforce. For example, if an individual has a higher life expectancy, they could experience a lower physical and psychological cost of work, meaning that they are able to work for a longer period of time. However, a counterexample to this could be that individuals with higher life expectancy earn and save enough money so that they are able to retire early, and therefore, there is no need to continue to be in the workforce.

The regression equations go as follows:

$$LFP_{ct}^{55-65} = \beta_0 + \beta_1 Avg_Life_Expec_{ct} + \mathbf{X}'_{ct}\Gamma + \gamma_t + \epsilon_{ct} \tag{1}$$

$$EMP_{ct}^{55-65} = \beta_0 + \beta_1 Avg_Life_Expec_{ct} + \mathbf{X}'_{ct}\Gamma + \gamma_t + \epsilon_{ct} \tag{2}$$

$$LFP_{ct}^{66-74} = \beta_0 + \beta_1 Avg_Life_Expec_{ct} + \mathbf{X}'_{ct}\Gamma + \gamma_t + \epsilon_{ct} \tag{3}$$

$$EMP_{ct}^{66-74} = \beta_0 + \beta_1 Avg_Life_Expec_{ct} + \mathbf{X}'_{ct}\Gamma + \gamma_t + \epsilon_{ct} \tag{4}$$

Where X'_{ct} is a vector of demographic and policy controls including:

- $ShareBA_{ct}$: Share of the population with a Bachelor's degree or higher.
- $ShareFemale_{ct}$: Share of county that is female.
- $ShareWhite_{ct}$: Share of county that is white.
- $GovHealthSpend_{ct}$: Average Government health expenditure by state per capita.
- $StateRetirementAge_{ct}$: Average retirement age by state.

5 Results

Table 1: 55–66 Labor Force Participation

	Estimate	Std. Error	t value	p-value	
Avg_Life_Expecl	0.00720558	0.00373236	1.930568	0.05420	.
Share_ba	0.32341325	0.03577552	9.040072	$< 2.2 \times 10^{-16}$	***
Share_female	-0.92880190	0.21497314	-4.320549	0.0000194	***
Share_white	0.04992412	0.01881120	2.653957	0.00825	**
Government_Healthcare	0.00000129	0.00000188	0.687343	0.49224	
State_Retirement_Age	0.00906104	0.00269181	3.366154	0.00083105	***

Table 2: 55–65 Employment Rate

	Estimate	Std. Error	t value	p-value	
Avg_Life_Expec	0.00585464	0.00400991	1.460043	0.14501	
Share_ba	0.33092288	0.03861672	8.569419	$< 2.2 \times 10^{-16}$	***
Share_female	-0.96005164	0.22160164	-4.332331	0.0000184	***
Share_white	0.06082104	0.02029041	2.997527	0.00288	**
government_Healthcare	0.000000115	0.00000198	0.057989	0.95378	
State_Retirement_Age	0.00862683	0.00283032	3.048004	0.00245	**

Table 3: 66–74 Labor Force Participation

	Estimate	Std. Error	t value	p-value	
Avg_Life_Expec	-0.01036942	0.00377792	-2.744741	0.00631	**
Share_ba	0.30516007	0.02854592	10.690148	$< 2.2 \times 10^{-16}$	***
Share_female	0.08685709	0.14492447	0.599327	0.54927	
Share_white	-0.03065793	0.01681515	-1.823233	0.06897	.
Government_Healthcare	0.00000272	0.00000213	1.274976	0.20301	
State_Retirement_age	0.01580976	0.00274557	5.758270	1.62×10^{-8}	***

Table 4: 66–74 Employment Rate

	Estimate	Std. Error	t value	p-value	
Avg_Life_Expec	-0.01101685	0.00380725	-2.893651	0.0040	**
Share_ba	0.30143971	0.02882433	10.457823	$< 2.2 \times 10^{-16}$	***
Share_female	0.07728900	0.14339983	0.538976	0.5902	
Share_white	-0.02536835	0.01684985	-1.505553	0.1329	
Government_Healthcare	0.00000266	0.00000212	1.255372	0.2100	
State_Retirement_Age	0.01576761	0.00270516	5.828718	1.10×10^{-8}	***

Based on the outputs in the tables above, we can conclude that life expectancy can predict labor force participation and employment rate in the 66-74 year old age group, but not in the 55-66 year old age group. As shown in Table 1 and Table 2, we can see that Avg_Life_Expec is not statistically significant, but Share_white, Share_female, Share_BA, and State_Retirement_Age are statistically significant within the 55-65 age group. When looking at economic theory, this could be due to the fact that this age group is still within the typical working ages. However, when we turn to the 66-74 age group, we can see that average life expectancy, share_BA, and State_Retirement_Age are all statistically significant for both LFP and EMP. This means that for every one unit increase in life expectancy for people ages 66-74, the labor force participation rate decreases by 1%. This also means that for every one unit increase in life expectancy for people ages 66-74, the employment rate also decreases by 1%. When looking at these results in a real-world context, we can see that they line up with the existing literature. As life expectancy increases, people are more able and willing to retire, reducing both labor force participation and employment as fewer older adults need or choose to remain in the workforce. The share_BA also makes sense as people who have higher levels of education are able to stay in the workforce for longer, as their jobs are not physically demanding. In addition, if someone took the time to get a specific degree, they probably enjoy their job, and it is not just a source of income. To simplify into a more digestible real world context these results show that living longer does not necessarily mean working longer. In addition, contrast to popular belief the window of begin retired is actually much longer then expected. This, in turn, means that higher life expectancy could be putting more pressure on Social Security because people are living longer but working less in their late 60s and 70s.

6 Conclusion

This paper examined whether higher county-level life expectancy leads to greater labor force participation among older adults in the United States. Specifically, I tested whether living in a county with a higher life expectancy predicts employment for individuals aged 55–65 and 66–74 after controlling for local economic and demographic conditions. The results show that there is a relationship between life expectancy and employment rate among individuals aged 66–74, but there is no statistically significant relationship for individuals aged 55–65. However, we did find that for people ages 55-65, demographic factors (share_BA, share_Female, share_White) consistently explain more variation in employment rate and labor force participation than health alone.

These findings suggest that improved longevity does not necessarily translate into longer attachment to the labor market at older ages. These results are consistent with life-cycle labor supply theory, in which retirement decisions are driven more strongly by financial incentives, accumulated wealth, human capital, and local labor market conditions. In wealthier and more educated counties, older individuals may be better positioned to retire due to greater access to savings, pensions, and social security. Alternatively, in areas with fewer economic opportunities, improvements in health alone may not be enough to offset these weaker incentives. Our study did have limitations, and we were not able to account for absolutely everything. For future studies, it would be beneficial to include more in-depth analysis of education levels, which could uncover more potential trends. In addition, since our data was also limited to a 10 year period, any additional, more complete data would also definitely help improve the accuracy of our model, as we could account for more of the real population.

7 References

Aitken, Andrew (July 9, 2025). *Navigating the golden years: Making the labour market work for older workers: OECD Employment Outlook 2025*. OECD. URL: https://www.oecd.org/en/publications/oecd-employment-outlook-2025_194a947b-en/full-report/navigating-the-golden-years-making-the-labour-market-work-for-older-workers_2bc681ed.html (visited on 01/26/2026) (Go back).

CMS (2026). *Health Expenditures by State of Residence, 1991-2020*. URL: https://www.cms.gov/data-research/statistics-trends-and-reports/national-health-expenditure-data/state-residence (visited on 01/26/2026) (Go back).

Fowler, Emily (Sept. 23, 2025). *The Average Retirement Age in Your State, and How It Compares to Your Savings*. Yahoo Finance. URL: https://finance.yahoo.com/news/average-retirement-age-state-compares-135653510.html (visited on 01/26/2026) (Go back).

Health Care Expenditures per Capita by State of Residence — KFF State Health Facts (2026). URL: https://www.kff.org/state-health-policy-data/state-indicator/health-spending-per-capita/?currentTimeframe=1&sortModel=%7B%22colId%22:%22Location%22,%22sort%22:%22asc%22%7D (visited on 01/26/2026) (Go back).

Kalwij, Adriaan and Frederic Vermeulen (2008). "Health and labour force participation of older people in Europe: What do objective health indicators add to the analysis?" In: *Health Economics* 17.5. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/hec.1285. ISSN: 1099-1050. DOI: 10.1002/hec.1285. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/hec.1285 (visited on 01/26/2026) (Go back).

Leonesio, Michael, Benjamin Bridges, Robert Gesumaria, and Linda Del Bene (2026). *The Increasing Labor Force Participation of Older Workers and its Effect on the Income of the Aged*. Social Security Administration Research, Statistics, and Policy Analysis. URL: https://www.ssa.gov/policy/docs/ssb/v72n1/v72n1p59.html (visited on 01/26/2026) (Go back).

Personnel Management, U.S. Office of (2015). "Retirement Statistics & Trend Analysis: Fiscal Year 2015 to Fiscal Year 2019". In: (Go back).